

Physics A (H156, H556)

Exploring - Using X-rays ppq KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is **57**.
- The total number of marks may take into account some 'either/or' question choices.

- 1(a) Radiographers commonly use molecules containing fluorine F-18 as tracers in positron emission tomography (PET) scanning.

Fluorine has a proton number of 9.

F-18 decays to oxygen (O) by β^+ decay.

Write the equation for the decay of a nucleus of F-18 using nuclear notation.

[2]

- (b) The β^+ particle (positron) produced travels only a short distance in the patient before it meets an electron and is annihilated.

Calculate the wavelength λ of gamma photons produced.

$\lambda = \dots\dots\dots\text{m}$ [3]

- (c) X-rays and gamma-rays are produced by different physical processes.

Briefly describe both processes.

----- [2]

(d) F-18 has a half-life of 109.7 minutes.

Explain the advantage that this has for the patient but the disadvantage that this has for the radiographers.

[3]

2(a)

Describe, in terms of X-ray photons, the attenuation mechanism of Compton scattering.

[2]

(b) A parallel beam of X-rays is incident normally on a tissue as shown in Fig. 24. 1.

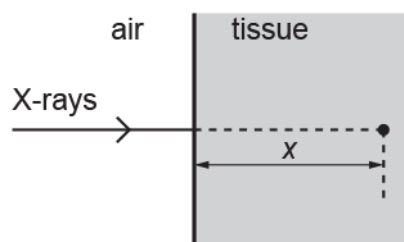


Fig. 24.1

The variation of the intensity I of the X-rays with depth x in the tissue is shown in Fig. 24. 2.

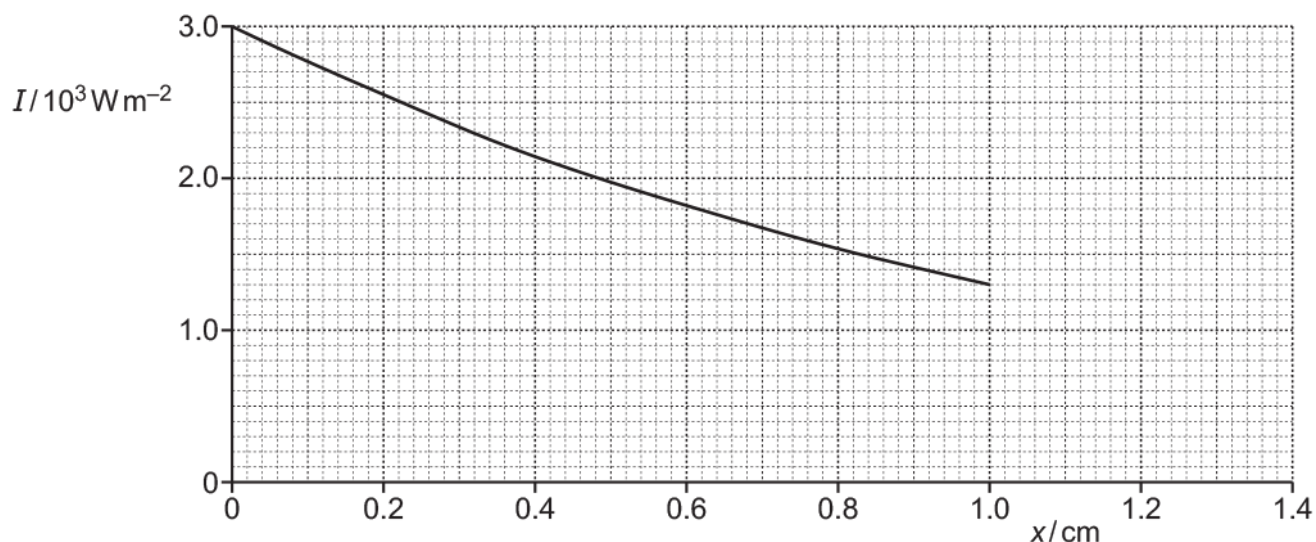


Fig. 24.2

The tissue has uniform structure between $x = 0$ and $x = 1.0 \text{ cm}$.

- i. Use the graph to determine the attenuation (absorption) coefficient μ in cm^{-1} of the tissue.

$$\mu = \dots\dots\dots \text{cm}^{-1} \text{ [2]}$$

- ii. Use the graph to determine the exposure time t for the total radiant energy incident per cm^2 at a depth of 1.0 cm to be 2.6 J .

$t = \dots\dots\dots$ s [3]

iii. Beyond $x = 1.0$ cm, the tissue has a larger attenuation coefficient than the value calculated in (i).

On Fig. 24.2, sketch the variation of I with x beyond $x = 1.0$ cm.

[2]

3(a) A high-energy X-ray photon interacts with an electron of an atom through the **Compton effect**.

Describe this effect.

[2]

(b) The diagram below shows a beam of X-rays incident normally on some soft tissue.

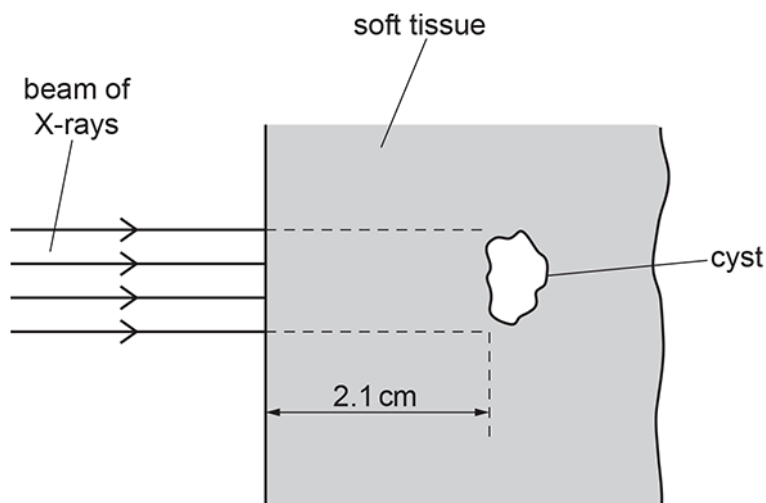


Fig. 2

The attenuation (absorption) constant of the soft tissue is 0.85 cm^{-1} .

The intensity of the beam is $4.6 \times 10^3 \text{ W m}^{-2}$.

There is a small cyst 2.1 cm from the surface of the soft tissue. The cross-sectional area of the cyst normal to the beam is $3.4 \times 10^{-4} \text{ m}^2$.

The beam is switched on for 30 s.

Calculate the X-ray energy incident on the cyst in a period of 30 s.

energy = J [4]

(c)

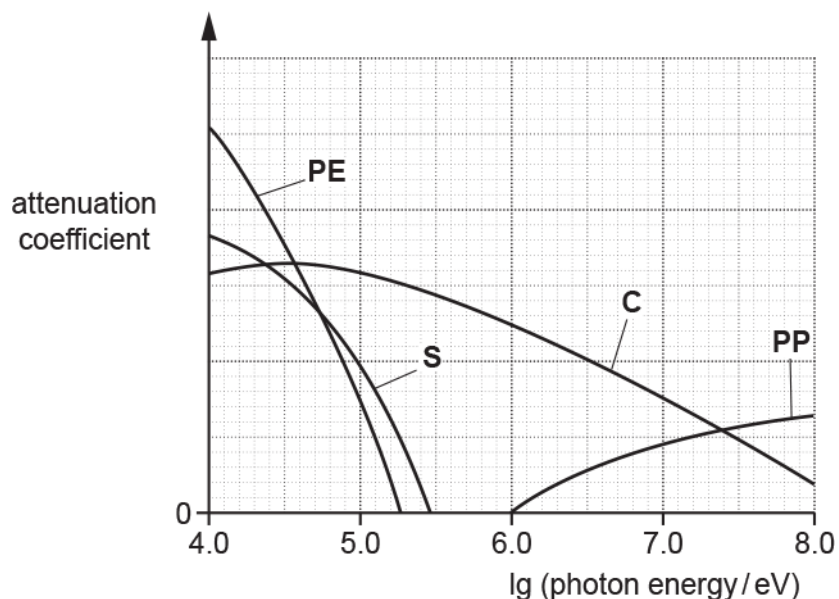
The attenuation coefficients of the cyst and the soft tissues in(b) were similar. This prevented imaging the cyst using a two-dimensional X-ray image.

Name a different X-ray technique that could be used to image the cyst. Explain the advantage of this technique.

[2]

4(a) X-ray photons interact with atoms.

The attenuation coefficient against $\lg(\text{photon energy})$ graphs for simple scattering (S), photoelectric effect (PE), Compton effect (C) and pair production (PP) are shown below.



For the X-ray tubes used in hospital, the X-ray photons have energy of about 10^5 eV.

State the attenuation mechanisms for these photons.

[1]

(b) With the help of a calculation, explain the minimum photon energy for pair production.

----- [3]

5 The Doppler effect is used to measure the speed of blood flow in arteries.

Which medical diagnostic method uses this technique?

- A barium scan
- B CAT scan
- C PET scan
- D ultrasound scan

Your answer ☐

[1]

You may draw a labelled diagram.

[3]

- (b) A beam of X-rays is directed at tissues in a patient.
The X-ray photons interact with the atoms of the tissues.

Simple scatter is one of the attenuation mechanisms.

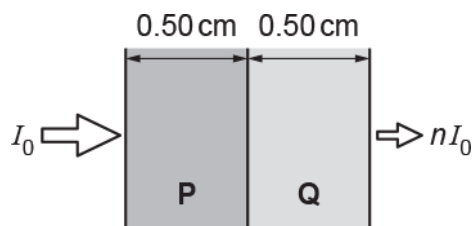
Name and describe **two** other attenuation mechanisms.

1

2

[4]

- 7 The intensity of a beam of X-rays incident on material **P** is I_0 .
The beam passes through 0.50 cm of material **P** and 0.50 cm of material **Q**.



The absorption (attenuation) coefficients of **P** and **Q** are 0.60 cm^{-1} and 0.20 cm^{-1} respectively.
The intensity of the beam after passing through both **P** and **Q** is nI_0 .

What is the value of n ?

- A 0.67
- B 0.74
- C 0.82
- D 0.90

Your answer

[1]

Describe how the components of a computerised axial tomography (CAT) scanner can produce high-quality images of the internal structures of a patient.

[41]

- A Iodine is a contrast material.
- B Technetium is a contrast material.
- C A contrast material must have a short half-life.
- D A contrast material is used for acoustic matching.

[1]

- 10 There are four important attenuation mechanisms by which X-ray photons may interact when they pass through matter.

In which mechanism is the X-ray photon scattered with a longer wavelength?

- A simple scattering
- B Compton effect
- C pair production
- D photoelectric effect

Your answer

[1]

- 11(a) Describe the X-ray attenuation mechanisms of *simple scatter* and *pair production*.

simple scatter

.....

.....

pair production

.....

.....

[2]

- (b) Calculate the maximum wavelength of the X-rays for the pair production process.

maximum wavelength = m [3]

- (c) An X-ray image of a patient's arm is required.

muscle

bone

muscle

X-ray beam

A

0 1 2 3 4 5

x / cm

X-rays with

The graph shows the intensity of light as a function of distance x in cm. The x-axis ranges from 0 to 5 cm with major ticks every 1 cm and minor ticks every 0.2 cm. The y-axis is labeled 'intensity' and has four major tick marks. The curve starts at a high intensity at $x=0$, decreases linearly to $x=3$, and then decreases more rapidly, approaching a constant value as x increases towards 5.

[4]

- 12 A radiographer in a hospital directs a parallel beam of X-rays at the leg bone of a patient. The attenuation (absorption) coefficient of bone is 0.7 cm^{-1} .

The answers below are given to one significant figure.

What is the percentage intensity of X-rays transmitted through bone of thickness 0.7 cm ?

- A. 0 %
- B. 40 %
- C. 50 %
- D. 60 %

Your answer

[1]

- 13 The potential difference across the cathode and the anode of an X-ray tube is V . The minimum wavelength of the X-ray photons emitted from the tube is λ_0 .

Which of the following statements is / are correct?

- 1. λ_0 is halved when V is doubled.
- 2. λ_0 is unchanged when the temperature of the cathode is increased.
- 3. λ_0 is independent of the cathode material.

- A. 1, 2 and 3
- B. Only 2 and 3
- C. Only 1 and 2
- D. Only 2

Your answer

[1]

END OF QUESTION PAPER

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
1	a	${}^{18}_{9}\text{F} \rightarrow {}^{18}_{8}\text{O} + {}^0_1\beta^+ (+\nu)$	C1 A1	Correct fluorine isotope Correct equation ALLOW numbers written to right of symbol ALLOW $e / e^+ / \bar{e} / \beta$ for positron symbol with correct numbers IGNORE gamma in products ALLOW ${}^0_0\nu$ for neutrino ALLOW ν_e for neutrino <u>Examiner's Comments</u> Most candidates will have gained the first mark for the correct isotope of fluorine. Those who didn't likely reversed the positions of the nucleon and proton number. It was noticeable that a large number of candidates were unable to balance the equation and while they had the correct values for the positron, the oxygen isotope was incorrect.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	b	$E = mc^2 = 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2$ $= 8.2 \times 10^{-14} \text{ (J)}$ $E = hc / \lambda = 8.2 \times 10^{-14} = 6.63 \times 10^{-34} \times 3 \times 10^8 / \lambda$ $\lambda = 3.00 \times 10^8 \div 1.2367 \times 10^{20} = 2.4 \times 10^{-12} \text{ (m)}$	C1 C1 A1	Working leading to evaluation of E ALLOW factor of 2 leading to 1.6×10^{-13} (J) NOT use of $\lambda = h/mv$ where v is 3×10^8 alone XP ALLOW use of $\lambda = hc/mc^2$ or $\lambda = h/mc$ ALLOW Use of electron mass = 0.511 MeV giving 8.2×10^{-14} (J) Correct to at least 2 significant figures Answer of 4.9×10^{-12} or 1.2×10^{-12} allow 2/3 due to incorrect accounting for factor of 2 <u>Examiner's Comments</u> Well over half of the candidates were able to score 2 or more marks on this question. A great deal scored 2 marks due to the inclusion of a factor of two, which was then not removed (or vice versa). Some candidates used the formula for kinetic energy – which by coincidence may give the correct answer – however this was not credited, nor was the more common use of the de Broglie formula. Examiners sometimes had a difficult decision on this question, whether the response was a physics error or not, and if in doubt a candidate would be awarded the marks.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	c	X-rays formed when <u>electrons</u> (in an atom) de-excite Gamma rays come from the decay of <u>nuclei</u> (in unstable isotopes)	B1 B1	ALLOW X-rays may be produced by acceleration / deceleration of (fast moving) electrons / X-rays are produced when (fast moving) electrons are incident on a metal target / X-rays may be produced when electrons lose kinetic energy ALLOW gamma rays come from (the decay of) radioactive <u>nuclei</u> / gamma rays come from the <u>nucleus</u> of unstable atoms / gamma rays come from the de-excitation of <u>nuclei</u> / gamma rays come from <u>annihilation</u> of particle-antiparticle (pairs) / IGNORE gamma from fission <u>Examiner's Comments</u> The main principle behind this question was to distinguish between the X-ray production by electrons and gamma ray production by nuclei. Fairly specific details were required for each, although a wide range of answers were accepted. This question discriminated well among the candidates with roughly equal fractions getting 0, 1 and 2 marks.
	d	The half-life is short Advantage: Exposure of the patient to <u>ionising</u> radiation is kept as low as possible. Disadvantage: (Radiographers have a) short time to scan/diagnose the patient	B1 B1 B1	ALLOW activity is high IGNORE it decays quickly ALLOW F18 has to be manufactured on site before use / high activity means exposure is high during handling IGNORE short time to treat the patient <u>Examiner's Comments</u> Although apparently a simple set of ideas, the detail required for this question meant that only a small fraction of candidates were able to gain full marks. It was necessary to state that the half-life was (relatively) short, which many of the stronger candidates did not do. The advantage to the patient requires the use of the term 'ionising' as this is the fundamental issue with the radiation. Similarly, candidates had to make it clear that the disadvantage was due to the short time to carry out the scan rather than just a statement of needing to work quickly.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
			Total	10	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
2	a	Photon scattered with less energy / X-ray of lower frequency / X-ray of longer wavelength	B1	Allow X-ray for photon and vice versa Ignore wave
		Electron removed (from the atom)	B1	Not electron scattered or electron removed from surface <u>Examiner's Comments</u> The Compton effect is part of the X-ray attenuation section of the specification 6.5.1(d). It appeared that some candidates were unfamiliar with this process and attempted to describe the photoelectric effect. There was a noticeable use of casual language in the description of the electron being removed from the atom and several candidates described the electron as moving up energy levels. Those who were able to obtain both marks generally gave clear and succinct responses.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	b	i	Identification from graph of two intensities and corresponding separation between them (x). Correct substitution into $I = I_0 e^{-\mu x}$ and fully correct calculation leading to $\mu = 0.84 \text{ (cm}^{-1}\text{)}$	C1 A1	Note check value of intensities to $\pm \frac{1}{2}$ small square. x missing implies $x = 1$. Graph misread lose all marks. May be seen from substitution into $I = I_0 e^{-\mu x}$. No POT error at this point Note I_0 and I must be in correct position in equation, so that ($I < I_0$). If I and I_0 are reversed, μ will be negative. Penalise this mark for negative answer, or working that would lead to negative answer. Penalise POT error here. Allow small range of values around 0.84 for variations in graph readings. Examiner's Comments The treatment of the logs was well done here (again, as with 22bi) and most of the candidates chose 0.0 and 1.0 cm to produce their values. The factor of 10^3 in the intensity was not needed for this calculation, so candidates who chose to ignore it (or did not see it) were unlikely to be penalised. The negative sign caused a problem if I and I_0 were reversed and candidates deliberately ignoring it would not score the second mark. There was a common alternative to using the 0.0 and 1.0 cm points where candidates used the "half-thickness" value at approximately 0.83 cm which would produce the same answer. As the graph scale on the vertical axis was simple, values which could have been from a "misread" would not be given further marks. Although readings were taken from a graph in the calculation, several candidates gave their final answer to 1sf despite calculating it correctly throughout.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	$I = 1.3 \times 10^3 / 1300$ $(E = I \times A \times t) = 2.6 = 1300 \times 1.0 \times 10^{-4} \times t$ $t = 20 \text{ (s)}$	B1 C1 A1	Allow a value in the range $1.25 - 1.35 \times 10^3$. Allow ecf from incorrect reading of I at 1.0cm from (b)(i) Allow use of value of I from B1 except use of 3000 POT error allowed on area if area clearly included in the calculation, or if A included in equation Note: use of $I = 1.6 \times 10^3$ leading to $t = 16 \text{ (s)}$ with A included scores 2 marks <u>Examiner's Comments</u> Many candidates made a good attempt at this question, however the conversion between cm^2 to m^2 and missing the 10^3 on the intensity reading caused two potential power of ten errors.
		iii	A curve starting at $x = 1.0\text{cm}$ with initially larger negative gradient at $x = 1.0 \text{ cm}$ Exponential curve	M1 A1	Must not have a positive gradient for more than half a square vertically Curve must not touch x-axis or be horizontal for more than 3 small squares <u>Examiner's Comments</u> Most candidates appreciated that a larger attenuation coefficient would lead to a line which would lead to a lower intensity than a continuation of the original line. Common (and understandable) errors included the line touching the x axis and a continuation of the original line of increasing gradient. Some candidates may have misinterpreted the question as they drew a second line from the (3.0, 0.0) point beneath the original line.
			Total	9	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
3	a	Electron removed / ejected (from atom)	B1	Needs a comparative statement
		Photon (scattered with) increased wavelength / lower frequency / lower energy	B1	<u>Examiner's Comments</u> While many candidates were aware of the Compton effect and were able to give some description, the lack of use of appropriate scientific terminology meant that some were unable to score any marks. The first marking point requires it to be clear that an electron is removed from the atom. A significant number of candidates described this in terms of moving shells, or excitation. The second marking point required the candidate to express that the photon had a lower energy (or equivalent) which again was often answered carelessly. Despite some good general descriptions, over half the candidates did not achieve a mark on this question.  Misconception As the Compton effect is further evidence for the particulate nature of light, it is important that this description is given in terms of photons. Several candidates gave an otherwise good description but gave it in terms of waves.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	b	$(\text{intensity } I = I_0 e^{-\mu x}) = 4.6 \times 10^3 \times e^{-0.85 \times 2.1}$ Either: (power =) $4.6 \times 10^3 \times e^{-0.85 \times 2.1} \times 3.4 \times 10^{-4}$ Or (energy per unit area =) $4.6 \times 10^3 \times e^{-0.85 \times 2.1} \times 30$ energy = $4.6 \times 10^3 \times e^{-0.85 \times 2.1} \times 3.4 \times 10^{-4} \times 30$ energy = 7.9 (J)	C1 C1 C1 A1	intensity = 772 (W m^{-2}) power = 0.262 (W) energy per unit area = 23160 J m^{-2} energy at surface = 47 (J) 2 marks <u>Examiner's Comments</u> There were many routes to a final answer in this question. Those candidates who set out their working carefully, used letters to represent the calculated quantity, and set this out in several stages tended to be the most successful. Some calculated the energy at the surface before going on to apply the attenuation formula, and others carried out the attenuation on the intensity. Each method can be credited at various stages, but it is important that a clear structure is shown. Many candidates attempted to change cm^{-1} to m^{-1} by dividing by 100, whereas the better candidates appreciated that the units of distance and attenuation constant would cancel in the exponent. Several candidates used the incorrect formula energy = power / time which can be a common misconception. The correct formula is in the data booklet if required.
	c	CAT (CT) scan Any <u>one</u> from • A CAT scan will give 3D image • A CAT scan gives better contrast	M1 A1	Insufficient: more detail / clearer image <u>Examiner's Comments</u> Many candidates used a technique that did not use X-rays (such as ultrasound) so could score no marks. Those that correctly identified the CAT scan did, in general, identify a correct advantage. Answers which were vague (along the lines of clearer) could not be credited. This question is directly from the learning outcome 6.5.1(g)
		Total	8	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4	a		All except pair production / PP	B1	Allow PE, S and C
	b		(energy =) $9.11 \times 10^{-31} \times (3.0 \times 10^8)^2$ (energy =) $2 \times 9.11 \times 10^{-31} \times (3.0 \times 10^8)^2 / 1.60 \times 10^{-19}$ $\lg 1.0(2) \times 10^6 = 6$ (as on graph) OR (energy =) 1.0×10^6 (eV) or $\lg 1.0 \times 10^6 = 6$ (from graph) (energy =) 1.6×10^{-13} J and evidence of mc^2 $2 \times 9.11 \times 10^{-31} \times (3.0 \times 10^8)^2 \approx 1.6 \times 10^{-13}$	B1 B1 B1 B1 B1 B1	Note this is 8.2×10^{-14} (J) Note this is $1.0(2) \times 10^6$ eV Note this can be shown in a variety of ways
			Total	4	
5			D	1	
			Total	1	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
6	a	(Evacuated tube with) cathode / heater / filament and target / metal / anode	B1	Ignore polarity of high voltage supply throughout
		High voltage (supply) connected between cathode and anode	B1	Note the first two B1 marks can be scored on a labelled diagram
		(Accelerated) electrons hit the target / metal / anode and their KE is transformed into X-ray (photons)	B1	Allow a specific value in the range 10 kV to 1 MV Note expecting 'high' or qualified by values in range above <u>Examiner's Comments</u> This question comes directly from the requirements of the learning outcomes 6.5.1(a) and 6.5.1(b) of the specification. In the first outcome, the basic structure of the X-ray tube outlines the components as heater (cathode), anode, target metal and high-voltage supply. Candidates generally gave poor answers with some components missed out, and quite often getting the anode and cathode very mixed up. On many scripts a cell or a battery was being used, instead of a high-voltage supply. In contrast some candidates incorrectly had a high-voltage supply connected to the heater in order to 'supply fast moving electrons released through thermionic emission'. There were missed opportunities with the explanation of the production of X-rays. Candidates did use terms such as Bremsstrahlung, but basic notion that the kinetic energy of the electrons was being used in the production of the X-ray photons eluded many candidates.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	b	Any two from: Photoelectric (effect) Photon (is absorbed and an) electron removed (from the atom) Compton (scattering / effect) Photon scattered / deflection with longer wavelength / low frequency / low energy and electron removed (from the atom) Pair production Photon (absorbed) and produce electron-positron (pair)	M1 A1 M1 A1 M1 A1	Ignore if interaction between photons and electrons is not one-to-one Not X-ray wave etc; must refer to photon Not reflected Not X-ray wave etc; must refer to photon Not X-ray wave etc; must refer to photon <u>Examiner's Comments</u> Most candidates correctly recalled the names of the remaining attenuation mechanisms and gave good descriptions. The mark scheme required descriptions in terms of the X-ray photons. The Compton effect was the most complicated mechanism, but many answers correctly described the scattering of the photon with reduced energy (or longer wavelength). A small number of candidates wrote about pair production as if it was the annihilation of an electron-positron pair. The majority of the candidates scored some marks for their answers.
		Total	7	
7		A	1	
		Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
8			X-ray (tube) moves around the patient A thin (fan-shaped X-ray) beam is used (Images / scans of) cross-sections through the patient are taken Any <u>one</u> from: • A three-dimensional image is produced • (Soft) tissues can be identified	B1 B1 B1 B1	Allow 'X-rays passed through different angles.' Allow 'slice(s)' Allow 'good contrast image' <u>Examiner's Comments</u> The majority of the candidates gave decent answers here for the CAT scanner. Answers were often very detailed and demonstrated a good understanding of the stages and the components used in the production of 3D X-rays images. Some even managed to give additional details on the absorption mechanisms. However, examiners only required simplistic answers in terms of a thin beam of rotating X-rays, which enabled cross-sectional scans (slices) of the patient and how computer software was used to generate 3D image of the patient. A small number of candidates outlined hybrid scanners with mixture of MRI scanners, gamma camera and PET scanners. Most of descriptions here earned the mark just for '3D image'. The omission rate for this last question was small enough for examiners to confidently say that there were no time issues with this H556/02 question paper.
			Total	4	
9			A	1	<u>Examiner's Comments</u> The majority of the candidates did get the correct answer A. A significant number of candidates opted for C, confusing contrast material with medical tracers.
			Total	1	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
10			B	1	
			Total	1	
11	a		Simple scatter: X-ray (photon) is scattered by an atomic electron. Pair production: An X-ray (photon) transforms into an electron and positron pair.	B1 B1	
	b		$\frac{hc}{\lambda} = 2 \times 9.11 \times 10^{-31} \times (3.0 \times 10^8)^2$ $\lambda = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{2 \times 9.11 \times 10^{-31} \times (3.0 \times 10^8)^2}$ $\lambda = 1.2 \times 10^{-12} \text{ (m)}$	C1 C1 A1	Allow 2 marks for $2.4 \times 10^{-12} \text{ (m)}$; factor of 2 omitted in the first line.
	c		The intensity decreases with thickness of muscle / bone. The decrease is exponential. The attenuation (absorption) coefficient μ of bone must be greater than the μ of muscle because there is a significant decrease in the intensity from $x = 3.0 \text{ cm}$ to 4.0 cm.	B1 B1 B1 B1	
			Total	9	
12			D	1	
			Total	1	
13			A	1	
			Total	1	